

ABDULRAHMAN ALSAADI

MECHATRONICS ENGINEER | MECHANICAL DESIGN

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PROFESSIONAL SUMMARY

CSWP-certified Mechatronics Engineer focused on mechanical design and electromechanical product development, with hands-on experience taking systems from CAD and engineering analysis through prototyping, manufacturing, assembly, physical testing, and iterative redesign. Experienced in SolidWorks, FEA, DFM, dimensional tolerancing, fits and clearances, transmission design, BOM development, and additive manufacturing.

TECHNICAL SKILLS

Mechanical Design & CAD: SolidWorks, Mechanism Design, Dimensional Tolerancing, Fits & Clearances, Bearing Fits, Manufacturing Drawings.

Analysis & Optimization: SolidWorks Simulation, FEA, Topology Optimization, Torque Analysis, DFM

Manufacturing & Documentation: BOM Development, 3D Printing, Prototyping, Technical Documentation, Assembly & Fit Validation

Electromechanical Systems: Belt & Gear Transmissions, Stepper/Servo/BLDC Motors, ESP32, Arduino, Raspberry Pi 5

SELECTED MECHANICAL DESIGN PROJECTS

SCARA ROBOTIC ARM - GRADUATION PROJECT

Karabük, Türkiye

Mechatronics & Robotics Engineer

Apr 2026 - Jun 2026

- Independently designed, manufactured, assembled, and physically validated a 3-DOF SCARA robotic arm in SolidWorks with a verified 492.5 mm reach and 100 g payload at full extension; developed custom links, bearing housings, motor interfaces, and transmission components.
- Applied DFM, dimensional tolerancing, fits and clearances, and bearing-fit design; redesigned components after identifying physical fit and assembly issues during prototyping, and produced the BOM and manufacturing drawings.
- Applied SolidWorks Simulation, FEA, and topology optimization to redesign Arm Link 1, reducing mass by approximately 16.5% while retaining critical mounting and load-bearing regions.
- Designed and integrated a two-stage GT2 belt transmission, NEMA 17 motor system, bearings, and servo gripper; performed torque calculations and manufactured functional PLA+ components using FDM 3D printing with drilling, reaming, tapping, and heat-set inserts.

WD DRONE TEAM - TEKNOFEST 2026 Competition, Rotary Wing Category

Karabük, Türkiye

Simulation & Systems Integration

Feb 2026 - Jul 2026

- Contributed to UAV mechanical development through component placement, BOM development, prototyping, assembly, fastener selection, and iterative redesign to resolve fit, assembly, and integration issues.
- Performed SolidWorks Simulation structural FEA on 5 mm carbon-fiber arm plates and 3 mm central-body plates under an 8 kg static load using supplier material properties; supported structural verification and component-design decisions.
- Supported manufacturing, mechanical assembly, and staged physical validation of the UAV, including an 8 kg structural load test, 12-minute unloaded hover, 7-minute hover with a 1 kg payload, and five flight-test sessions.

6-DOF CYCLOIDAL ROBOTIC ARM – PERSONAL PROJECT

Jeddah, Saudi Arabia

Mechanical Design

Sept 2026 - Sept 2026

- Designed a 6-DOF robotic arm in SolidWorks with five custom single-stage 27:1 cycloidal-reducer joints, targeting a 450 mm reach and 0.6 kg payload; defined the kinematic architecture and performed joint torque, inertia, and motor-sizing calculations.
- Developed the mechanical architecture from component-level design through bearing and shaft selection, fits and tolerances, reducer packaging, DFMA, BOM development, and manufacturing drawings; designed dual-bearing joint supports and a synchronized parallel gripper for future prototyping and fabrication.

EXPERIENCE

SMART METHODS

Jeddah, Saudi Arabia

Full Stack Robotics Engineering Intern

Jul 2025 - Aug 2025

- Designed a complete 5-DOF robotic-arm concept in SolidWorks, including links, joint structures, motor mounts, and overall mechanical architecture.
- Redesigned a robotic-arm joint bracket with an integrated bearing housing to reduce unnecessary material while maintaining functional requirements; performed SolidWorks Simulation FEA using ABS material and 100 N loading per edge, obtaining a calculated factor of safety of approximately 22.5.
- Performed static torque calculations and servo-motor selection for a 3-DOF robotic arm designed around a 1 kg payload, relating link geometry and gravitational loading to actuator requirements.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering

Jun 2021 - Jun 2026

Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- Certified SOLIDWORKS Professional - Mechanical Design (CSWP), Dassault Systèmes | Aug 2026
- Saudi Council of Engineers (SCE) - Engineer Membership (Active)
- Certified SOLIDWORKS Associate — Mechanical Design (CSWA), Dassault Systèmes | May 2026

LANGUAGES

Arabic - Native | English - Upper-intermediate | Turkish - Intermediate